

ChainMark

A Cryptographic Breadcrumb Architecture for CSAM Distribution Network Interdiction

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Abstract

Existing tools for combating child sexual abuse material (CSAM) — primarily perceptual hash databases such as PhotoDNA — are effective at identifying known content but are architecturally passive: they detect individual files in isolation and cannot reconstruct distribution networks. This proposal introduces ChainMark, a law-enforcement-controlled cryptographic watermarking and payload-mutation architecture designed for use in authorized sting operations. ChainMark embeds a uniquely keyed, steganographic breadcrumb into seeded files that mutates at each forward event, producing a self-documenting chain-of-custody log that maps the topology of a distribution network in real time. Entire networks can be identified and interdicted simultaneously rather than sequentially. The system is designed with a strict key-custody model ensuring only authorized law enforcement personnel can read or generate payloads.

1. Problem Statement

Current federal strategy against CSAM distribution networks faces a structural limitation: arrests are made sequentially, one node at a time. When a single actor is identified and apprehended, the remainder of the network typically disperses, migrates platforms, or rotates keys before investigators can act. The result is a whack-a-mole enforcement cycle that imposes real operational cost without achieving network-level disruption.

PhotoDNA and NCMEC hash-matching are valuable for detecting known content at scale, but they answer the question 'is this file known illegal content?' — not 'who is in possession of it, who gave it to them, and who have they given it to?' The investigative leap from file detection to network topology currently requires extensive manual subpoena chains, cooperative platform disclosure, and

sequential legal process — each step giving the network time to adapt.

ChainMark addresses this gap. Rather than waiting for content to appear in hash databases, it makes the content itself the investigative instrument.

2. Prior Art and Differentiation

System	Function	Network Mapping?	Limitations
PhotoDNA (Microsoft/NCMEC)	Perceptual hash matching of known files	No	Passive; identifies files, not actors or chains
NCMEC CyberTipline	Platform-reported hash matches	No	Requires platform cooperation; no propagation trace
Traditional Watermarking	Static ID embedded in media	Partial (single hop)	Payload does not mutate; cannot trace multi-hop chains
FBI Sting Ops (existing)	Undercover distribution	Partial	Manual, resource-intensive; not scalable to full network
ChainMark (proposed)	Mutating cryptographic breadcrumb chain	Yes — full topology	Requires seeded origin; cannot passively infect existing networks

3. ChainMark Architecture

3.1 Core Concept

ChainMark operates on a controlled-seeding model. Law enforcement, acting under lawful authority, introduces a specially prepared file into a target network. The file contains a steganographic payload invisible to human observers and resilient to standard re-encoding operations. Unlike static watermarks, the ChainMark payload is designed to mutate at each distribution event — each time the file is forwarded, the receiving client (if running a ChainMark-aware relay, or via a paired device-level mechanism in warranted scenarios) appends a signed event record before passing the file along.

3.2 Payload Structure

Each ChainMark payload consists of three components embedded via spread-spectrum steganography:

Component	Description	Size (approx.)
Origin Token	Signed identifier of the seeding operation, encrypted under FBI master key	256 bits

Event Chain Log	Append-only list of signed forward events; each event: timestamp + node identifier hash	Variable (grows per hop)
Integrity MAC	HMAC over full payload; detects tampering; recalculated at each mutation event	256 bits

3.3 Mutation Protocol

The mutation mechanism is the critical innovation. On each distribution event, the following sequence executes at the relay layer (whether that is a warranted device agent, a cooperating platform integration, or an undercover-operated relay node):

Step 1 — Extract: The current payload is extracted from the steganographic channel using the ChainMark read key. Step 2 — Append: A new event record is constructed: {timestamp, node_id_hash, predecessor_MAC}. Step 3 — Re-sign: The updated event chain is signed with the relay node's ephemeral key (issued by FBI PKI). Step 4 — Re-embed: The mutated payload is embedded back into the file via spread-spectrum steganography. Step 5 — Forward: The modified file is passed to the next node. The relay node logs the event to the central ChainMark registry.

CHAIN RECONSTRUCTION: At any point, law enforcement can extract the full event chain from any recovered file, producing a complete, cryptographically signed map of every node the file passed through — in order, with timestamps. This enables simultaneous warrants across all identified nodes.

3.4 Steganographic Robustness

The payload must survive common file transformations that occur during distribution: JPEG re-compression, color space conversion, minor cropping, and scaling. ChainMark employs a spread-spectrum embedding strategy in the DCT coefficient domain for JPEG/video content, distributing payload bits across the frequency spectrum in a pattern keyed to the master secret. This approach is robust to compression up to approximately 60% quality reduction and survives 5-10% spatial cropping. For higher-destruction scenarios, a redundant embedding in multiple frequency bands provides additional resilience.

4. Key Custody and Access Control

The security of ChainMark against misuse is entirely dependent on strict key custody. The architecture enforces a three-tier key hierarchy:

Tier	Key Type	Held By	Function
Tier 1	Master Read/Write Key	FBI CEAU (sole custody) Hardware Security Module	Generates seeding keys; can read any payload
Tier 2	Operation Keys	Assigned per operation; custody logged in HSM	Authorizes a specific sting operation seed

Tier 3	Node Relay Keys	Issued ephemerally per warranted device	Signs individual mutation events only
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No entity below Tier 1 can read payload content generated by another operation. Node relay keys expire after 72 hours and are non-renewable without Tier 1 authorization. All key generation and usage events are immutably logged. The master key never leaves the HSM.

4.1 Operational Lifecycle

Key lifecycle follows a strict authorization-to-retirement sequence. Before any seeding operation is authorized, a Tier 1 officer generates a unique Operation Key scoped to that warrant and that warrant alone. The operation key is cryptographically bound to the warrant number at generation time — it cannot be repurposed or reused across operations. When the operation concludes, the operation key is formally retired in the HSM log, and all Tier 3 node relay keys issued under it are simultaneously invalidated. A retired key cannot be reinstated. The full lifecycle — generation, issuance, use, and retirement — produces an immutable HSM log entry at every stage, providing a complete audit trail that can be produced in discovery without exposing key material.

4.2 Hardware Security Module Requirements

The Tier 1 master key must reside exclusively in a FIPS 140-2 Level 3 certified Hardware Security Module or higher. This certification level requires physical tamper resistance: any attempt to extract key material from the device destroys the key. The HSM must be operated in a physically secured facility under dual-person integrity — no single officer can authorize or execute a master key operation alone. The HSM audit log must be append-only, cryptographically signed, and replicated to an offline backup at minimum every 24 hours. These requirements are not aspirational — they are the minimum bar for ChainMark evidence to survive a Daubert challenge on the integrity of the key custody system.

4.3 Compartmentalization Principle

Operation Keys are compartmentalized: an officer with custody of an Operation Key for Case A cannot read, seed, or extract payloads from Case B, even if they hold Tier 2 authorization for both. This compartmentalization serves two purposes. Operationally, it limits blast radius if a single Operation Key is compromised — no other active operation is exposed. Legally, it ensures that the evidentiary record for each case is cryptographically isolated, preventing any argument that evidence from one investigation contaminated or influenced another.

5. Evidentiary and Legal Considerations

A system that produces technically sound output but fails to meet courtroom evidentiary standards has no operational value. This section addresses the legal architecture ChainMark must satisfy to produce prosecutable evidence rather than mere intelligence.

5.1 Evidence Must Survive Court, Not Just Exist

The chain ChainMark produces is only useful if a prosecutor can make the following statement without qualification: this chain proves this specific person distributed this file. That requires three things the current architecture must explicitly address. First, every hop must be attributable to a real actor, not just a device hash. Second, the reconstruction method must be repeatable and explainable to a jury without specialist knowledge. Third, there must be no opaque steps: any black-box element in the reconstruction chain is an attack surface for the defense. If any single step cannot be explained plainly, defense counsel will argue the entire chain is unreliable.

5.2 Non-Manufacture Position

The most dangerous legal tripwire for ChainMark is the entrapment defense. The moment a system introduces content into a network and tracks its spread, defense will argue the government created the crime. ChainMark's legal framing must be airtight on three points: the system tracks propagation and does not induce it; any subject who receives and forwards the file must demonstrate pre-existing predisposition independent of ChainMark; and the mechanism must not alter user intent or behavior in any way. The file must be indistinguishable from any other file the subject would have sought, received, and forwarded absent the operation. If any of these lines blur, the prosecution's case collapses before the technical evidence is heard.

NON-MANUFACTURE REQUIREMENT: DOJ CEOS review of the predisposition standard is not optional — it is a prerequisite for any deployment. This review must occur before any technical pilot is authorized.

5.3 Chain-of-Custody Narrative Simplicity

Courts prefer boring clarity over technical cleverness. The narrative a prosecutor needs is: this file was created here; it passed through these parties in this order; no one could have altered this record without detection. The technical depth in this proposal exists to satisfy expert witnesses and Daubert hearings — it must never reach a jury in raw form. ChainMark's output layer must be designed from the start to produce a human-readable, chronologically ordered chain-of-custody report that a non-technical juror can follow without assistance.

5.4 Failure Model: Partial Chains Must Still Hold

If part of the chain fails — air gap, adversarial stripping, absent relay — the rest must still produce usable evidence. Each hop must be individually provable without depending on full chain continuity. A two-hop confirmed chain is prosecutable for those two hops. A recovered file with only an origin token still proves possession of a law-enforcement-seeded file. All-or-nothing dependency is a

design flaw, not an acceptable limitation. The architecture must be built around independent hop provability from the start.

5.5 Identity Linkage

A chain of cryptographic hashes proves nothing in court without a verified bridge to a human being. The required linkage chain is: node identifier to device (MAC address, hardware fingerprint, or IP log); device to account (platform subpoena or device seizure); account to person (subscriber records, biometric access logs, or witness testimony). ChainMark produces the node identifier. The investigative steps that complete the chain to a person are conventional law enforcement work — but they must be planned and executed for every node before charges are filed. Intelligence without identity linkage is not evidence.

5.6 Abuse Prevention Must Be Explicit and Auditable

Key custody alone is not sufficient. Defense counsel during discovery will ask: what prevents unauthorized seeding; what prevents internal misuse by authorized personnel; what audit trail exists for every operation from authorization through recovery. The answers must be structural, not policy-based. Misuse must be not merely discouraged but cryptographically detectable and attributable. Every seed operation requires dual authorization logged in the HSM. Every key issuance is tied to a named officer and a specific warrant number. Every payload extraction produces a tamper-evident audit entry. Unauthorized use must leave a trail as clear as authorized use.

5.7 The Jury Explanation Standard

Every technical element of ChainMark must pass this test before deployment: can it be explained to a juror in approximately thirty seconds? The following is the proposed baseline jury explanation:

"Each time this file was shared, it added a hidden, signed receipt. Those receipts cannot be forged or removed without breaking the record. When we recovered the file, we could read every step it took — who had it, when, and who they gave it to."

Any component of ChainMark that cannot be reduced to language consistent with the above must be redesigned or moved to expert witness context rather than direct jury presentation. Complexity that cannot be explained will be weaponized by the defense.

6. Honest Assessment of Technical Limitations

This proposal is submitted in good faith and the author declines to overstate its capabilities.

Origin Dependency: ChainMark requires law enforcement to be the seeding origin of the tracked file. It cannot passively infect existing networks or retrofit tracking onto files already in circulation. This limits application to proactive sting operations, not reactive detection.

Relay Cooperation Requirement: For the mutation mechanism to function without a device-level agent, a cooperating relay — either an undercover-operated node or a platform integration — must

be present at each hop. Fully air-gapped, manual physical transfers break the chain. The static payload (origin token) still survives and provides value even when mutation fails.

Adversarial Stripping: A sufficiently sophisticated actor with knowledge of ChainMark's existence could strip steganographic payloads by aggressive transcoding (e.g., recompressing at very low quality, printing and rescanning). The system's effectiveness depends in part on targets not knowing the mechanism exists — standard operational security for law enforcement tools.

Legal Framework: Seeding operations raise entrapment considerations that require careful legal review. This is an existing challenge for sting operations generally and is not unique to ChainMark, but must be addressed by DOJ counsel before deployment.

Scope Boundary: ChainMark is a network-mapping tool, not a content-generation or content-distribution system. The nature of the seeded content and all operational decisions remain exclusively within law enforcement authority and legal process. This proposal covers the cryptographic and steganographic architecture only.

7. Reference Implementation Sketch

The following pseudocode illustrates the three core ChainMark operations. Each function maps directly to a discrete, auditable event in the operational lifecycle: the seed event (origin, Tier 1 authorized), the mutation event (each relay hop, Tier 3 authorized), and the recovery event (extraction and reconstruction, Tier 1 authorized). The pseudocode is intentionally abstracted from any specific steganographic library to remain implementation-agnostic. A full reference implementation in Python targeting standard spread-spectrum DCT embedding is available for review by authorized personnel upon request.

7.1 Payload Embedding (Seed Operation)

The seed operation is the origin event. It is called exactly once per operation, under Tier 1 authorization, and produces the seeded file that will enter the target network. The origin token is signed with the master key, binding it irrevocably to the authorizing operation. The event chain is empty at this stage — the seed is the root of the chain, not a node in it. The resulting HMAC covers both the origin token and the empty chain, establishing the integrity baseline against which all future mutations are verified.

```
# ChainMark seed – called once at operation origin def
chainmark_seed(image_bytes, operation_id, master_key): origin_token =
sign(operation_id, master_key) # Tier 1 key event_chain = [] # empty at origin
mac = hmac(origin_token + encode(event_chain), master_key) payload =
encode(origin_token, event_chain, mac) return
spread_spectrum_embed(image_bytes, payload, master_key)
```

7.2 Payload Mutation (Relay Operation)

The mutation operation executes at each authorized relay node — every point in the chain where the file passes through a ChainMark-aware component. It begins with a tamper verification: if the MAC does not validate, the operation halts and the anomaly is logged. On a clean validation, a new event record is constructed containing the current timestamp, a hash of the node identifier (not the raw identifier, preserving operational security), and the predecessor MAC — creating a cryptographic link to the prior state. Critically, the event is logged to the central registry independently of the file itself, meaning the chain record persists even if the file is later destroyed, stripped, or lost.

```
# Called at each authorized relay node def chainmark_mutate(image_bytes,
node_id, relay_key, read_key): payload = spread_spectrum_extract(image_bytes,
read_key) origin, chain, mac = decode(payload) assert verify_mac(origin, chain,
mac, read_key) # tamper check event = sign({'ts': now(), 'node': hash(node_id),
'prev': mac}, relay_key) chain.append(event) new_mac = hmac(origin +
encode(chain), relay_key) new_payload = encode(origin, chain, new_mac)
log_to_registry(operation_id=origin, event=event) # central log return
spread_spectrum_embed(image_bytes, new_payload, read_key)
```

7.3 Chain Extraction (Recovery Operation)

The recovery operation is called on any file seized from a target — at arrest, during a search, or from a device image. It requires only the master read key, which never left the HSM. The output is a structured network path: an ordered list of every node the file passed through, with timestamps, linked by cryptographic predecessor references. The `build_network_graph` call converts this list into a graph representation suitable for visualization and for inclusion in the chain-of-custody report produced for prosecution. Each node in the graph corresponds to a discrete distribution event — a human action, not an automated system artifact — which is the evidentiary unit that maps to a specific person once identity linkage is completed.

```
# Called on any recovered file to reconstruct the network path
def chainmark_recover(image_bytes, master_key):
    payload = spread_spectrum_extract(image_bytes, master_key)
    origin, chain, _ = decode(payload)
    operation_id = verify_and_decode(origin, master_key)
    network_path = [decode_event(e) for e in chain]
    return { 'operation': operation_id, 'path': network_path, # ordered list of nodes with timestamps
            'graph': build_network_graph(network_path) }
```

All three operations are designed to be independently verifiable by a defense-retained expert given the public specification of the embedding algorithm and the relevant public keys. The master key itself is never disclosed — only the signed output it produced. This separation means ChainMark evidence can be independently validated without compromising the key material or any other active operation.

8. Recommended Submission Path

This proposal is most appropriately directed to the following federal entities for review and potential pilot development:

FBI — Child Exploitation and Human Trafficking Unit (CEHTU) — Primary operational consumer; sting operation authority.

HSI — Cyber Crimes Center (C3) — Parallel jurisdiction; cross-border network investigations.

NCMEC — Technology Coalition — Integration with existing hash infrastructure and reporting pipeline.

DARPA / IARPA — R&D; funding pathway for technical maturation of the steganographic and cryptographic components.

DOJ — CEOS (Child Exploitation and Obscenity Section) — Legal framework review for sting operation compliance.

Given the technical maturity of the underlying components, DARPA and IARPA represent credible funding pathways for a formal pilot program. The FBI CEHTU and HSI C3 are the appropriate first

points of contact for operational review.

9. Conclusion

The technical foundation for ChainMark exists today. Spread-spectrum steganography, cryptographic signing, and append-only chain architectures are mature disciplines. What ChainMark proposes is their deliberate integration into a law-enforcement-controlled operational tool purpose-built for network-level interdiction of CSAM distribution rings.

The shift from sequential arrest to simultaneous network interdiction represents a qualitative change in enforcement capability. Sequential takedowns are operationally expensive and give networks time to adapt. Simultaneous network-wide action, made possible by a real-time chain-of-custody map, eliminates that window.

The author's role in this effort ends at the paper and any reference implementation made available for government review. Operational authority, legal authorization, and deployment decisions rest entirely with the relevant federal agencies. This proposal exists to put a viable technical option on the table for the people whose job it is to use it.